# GNN-JFL: Graph Neural Network for Video SAR Shadow Tracking With Joint Motion-Appearance Feature Learning

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***Abstract*— In this study, we address the challenges associated with video synthetic aperture radar (Video SAR) shadow tracking, a technique used for continuous monitoring of ground moving targets. Due to challenges such as changes in shadow appearance, low contrast between shadow and background, and scene occlusion in Video SAR, existing methods often encounter extensive matching errors in the data association process, resulting in unsatisfactory tracking performance. To overcome these issues, we propose a novel method, graph neural network – joint feature learning (GNN-JFL), which is based on joint motion-appearance feature extraction and graph neural data association. This method uses the detector as a flexible plugin and introduces two key improvements in the tracker section to enhance tracking accuracy. First, we introduce JFL to extract the complementary appearance and motion features from shadow shapes and positions, obtaining more robust feature representations to improve the tracking performance under intricate challenges. Second, by organically integrating multiobject tracking (MOT) problems and graph neural networks (GNNs), we propose a novel GNN-based shadow tracking architecture, which utilizes graph relationships to learn the associations between shadows for more accurate tracking predictions. Our method is validated using two measured datasets and demonstrates superior performance in terms of MOT accuracy (MOTA). It outperforms the suboptimal method by 4.2% and 3.6% in the two datasets. This research contributes to the advancement of continuous monitoring techniques employing Video SAR shadow tracking.**

***Index Terms*— Graph neural network (GNN), multishadow tracking, video synthetic aperture radar (Video SAR).**

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## I. Introduction

VIDEO synthetic aperture radar (Video SAR) is commonly used in the field of ground moving target monitoring, as it can provide continuous observation of areas of interest under all-day and all-weather conditions [1], [2]. The depicted positions of moving targets in Video SAR images often deviate from their actual locations [3], while the shadows cast by these targets tend to accurately represent their true positions. As a result, a significant body of research has been dedicated to the development of methodologies that utilize these shadow features for more accurate tracking of moving targets in Video SAR [4], [5], [6].

Traditional Video SAR moving target tracking methods mainly rely on manually designing the features of moving targets or combining relevant prior knowledge for tracking. For instance, Zhao et al. [7] considered the spatial–temporal context information jointly to enhance tracking performance, while Wen et al. [8] proposed a track-before-detect framework leveraging the shadow amplitude and Doppler energy of the target to determine its existence. However, these traditional methods, which heavily depend on expert experience in feature design and prior knowledge, often struggle with accuracy in complex or varying scenes, especially when tracking multiple targets. Efforts have been made to utilize the excellent feature learning ability of deep learning for Video SAR tracking, but these applications are typically restricted to single-target-tracking tasks. For instance, Yang et al. [9] employed the SiamFc network for Video SAR shadow tracking, and Bao et al. [10] proposed a guided anchor Siamese network (GASN) based on image-template similarity for tracking arbitrary targets in Video SAR. Despite their effectiveness, these mainly focus on single-object tracking and thus fall short in practical applications such as Video SAR surveillance that require multiobject tracking (MOT). Therefore, deep learning-based Video SAR multishadow tracking methods have become current research hotspots.

Recent years have seen a surge of deep MOT methods based on optical video [11], [12], [13], [14], [15], [16]. The promising results have subsequently inspired the development

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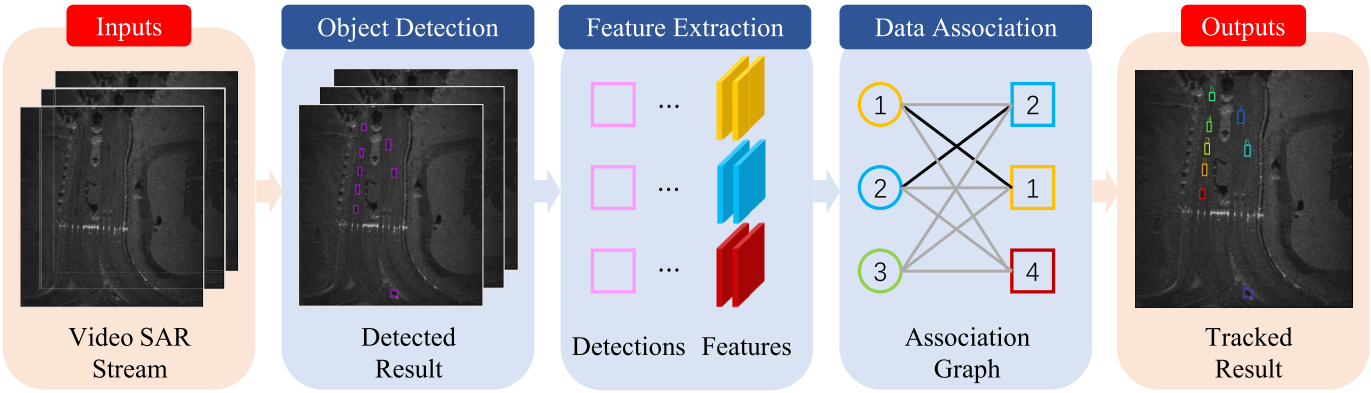



Fig. 1. Video SAR tracking process based on tracking-by-detection framework.

of Video SAR multishadow tracking field. Most existing deep MOT methods apply tracking-by-detection (TBD) pipelines [17], consisting of two parts: a detector and a tracker. Among them, the detector is employed to obtain target observations, such as the bounding box in the current scene. The tracker is responsible for associating the observed values obtained in the current frame with previously recognized objects and assigning different identities to multiple targets. Fig. 1 shows the process of using the TBD pipeline to achieve the Video SAR MOT task. The process starts by inputting the Video SAR image sequence into the detector frame by frame to obtain the detected results. It extracts the corresponding features from each detected object. Subsequently, the objects in the preceding and succeeding frames are matched based on feature similarity, thereby generating the final tracking results.

SAR images, while offering unique advantages, are often challenged by characteristics such as low resolution, strong background clutter, and speckle noise. This results in a quality disparity when compared to optical images [18]. Therefore, researchers have been innovating and adapting optical tracking methods for Video SAR shadow tracking. Inspired by FairMOT [15], Yang et al. [19] improved the MOT framework for SAR shadow tracking by using the triple attention (TriAtt) mechanism and lightweight multiscale network in the detector, which boosts tracking efficiency and performance. Its tracker part repeatedly utilizes the appearance features extracted by the backbone network to generate the re-identification [re-identity document (ID)] features for data association. Bao et al. [20] proposed a shadow-enhanced self-attention and anchor-adaptive network (SE-SA-AAN) based on SiamMOT [14] to achieve precise moving target tracking for Video SAR, reducing the serious false alarms and missed detections caused by optical tracking methods directly used in Video SAR through the self-attention and anchor-adaptive mechanism in the detecter. Its tracker part mainly utilizes the shadow appearance features extracted by the Siamese backbone network for correlation operation to achieve data association. Despite the competitive tracking results achieved, they mainly focus on improving detector performance, failing in addressing data association errors in the tracker. In Video SAR shadow tracking, the dynamic appearance of moving targets' shadows and the potential for shadows to disappear due to terrain and obstacles present unique challenges. By applying inconsistent shadow appearance features as the re-ID feature, current shadow tracking methods encounter many matching errors in the data association process. This leads to frequent switching of target identities and discontinuous tracking trajectories, which seriously deteriorates the final tracking performance.

Apart from detector accuracy, the tracking performance of TBD-based MOT method primarily depends on two aspects: the effectiveness of feature representation and the accuracy of matching algorithm based on the extracted features. To address the aforementioned challenges, we propose a novel method for Video SAR moving target tracking. This method, which we refer to as graph neural network – joint feature learning (GNN-JFL), is based on joint motion-appearance feature extraction and graph neural data association. Our method utilizes the detector as a flexible plugin and adopts the following two improvements in both feature extraction and data association aspects to improve tracking accuracy.

First, we design a joint feature extraction module to facilitate JFL. This module applies long short-term memory (LSTM) and convolutional neural networks (CNNs) to capture motion and appearance features from the trajectory and cropped images of the target detections, respectively. This allows us to integrate the motion features and appearance features of objects as the final feature representation. The motion features of shadows are not affected by shape changes, while the appearance features of shadows remain consistent in the short term. These are two complementary features. By comprehensively utilizing them as the feature representation of shadows, it helps reduce matching errors caused by occlusion and shadow changes, thereby enhancing the accuracy of data association and robustness against interference.

Second, we propose a novel GNN-based tracking network to make accurate tracking predictions. By utilizing detected shadows in consecutive frames as nodes, shadow features as node content, and the feature similarity between shadows as edges, a bipartite relationship graph can be constructed so that we convert the data association in MOT task into solving graph matching problem. Naturally, we leverage GNN for end-to-end learning to solve this graph matching problem. In comparison to traditional TBD-based tracking methods that directly use the extracted features for matching, GNNs optimize the relationship graph through feature interaction between nodes to obtain more discriminative features. Matching based

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on GNN-optimized features can avoid matching confusion, resulting in more accurate predictions for target associations. Specifically, we employ motion GNN and appearance GNN to optimize the extracted shadow features. We utilize the graph attention network (GAT) as the motion GNN and appearance GNN instead of the graph convolutional network (GCN), which utilizes GAT's more powerful feature aggregation mechanism to further enhance tracking accuracy. The main contributions of our method are as follows.

1) We establish a novel network GNN-JFL for Video SAR accurate moving target tracking. Specifically, two key improvements, joint motion-appearance feature extraction and graph neural data association, are proposed in the tracker section to enhance tracking accuracy.
2) We propose joint motion-appearance feature learning to extract more effective feature representations of targets, which comprehensively utilizes the appearance features and motion features of shadows as the final discriminative feature in Video SAR MOT. It leverages the complementary motion-appearance features to alleviate the instability of solely utilizing shadow appearance features, thereby enhancing the accuracy of tracking and providing a robust solution for the challenges posed by dynamic environments.
3) We propose graph neural data association to make accurate predictions, which transforms data association in MOT task into solving graph matching problem and naturally utilizes GNN to learn the association between shadows to obtain the optimal solution of graph matching problem. Specifically, we leverage the motion GNN and appearance GNN to optimize the extracted shadow features and utilize GAT to improve the accuracy of shadow matching.
4) We conduct extensive experiments on two datasets to confirm the effectiveness and robustness of the proposed method. The experimental Video SAR data are obtained from Sandia National Laboratories (SNL) and China Aerospace Science and Industry Corporation (CASIC) 23 research institute. The experimental results on both datasets demonstrate that GNN-JFL outperforms other six advanced methods.

The rest of this article is organized as follows. Section II presents the related works to our proposed tracking algorithm. Section III introduces the overall tracking framework, joint feature extraction, and GNN for data association in detail. Section IV shows the experimental results and analysis of the two Video SAR datasets. Section V concludes this article.

## II. Related Works

### A. Online Deep MOT

Due to the rapid processing of input video sequences, online deep MOT [21], [22] has been widely applied in real scenarios and has gradually become a hot research topic. The difference between online MOT and offline MOT is that the online tracking method processes the video frames as they arrive, without having access to future frames. Next, all the MOT methods mentioned in this article are used for online MOT tasks. In recent work, most deep MOT methods have adopted these two paradigms: TBD and joint detection tracking (JDT). The difference between these two paradigms is that the MOT method based on the TBD paradigm treats object detection and re-ID as two separate tasks and studies appropriate solutions for each task to achieve optimal performance. However, the MOT method based on the JDT paradigm combines the idea of multitasking learning using a single network to simultaneously complete object detection and identity embedding to reduce inference time. The subsequent data association process of these two paradigms is the same: calculate the affinity matrix based on the pairwise similarity of the learned re-ID features, construct a matching relationship graph, and use the Hungarian algorithm to solve the bipartite matching problem finally [23].

Thus, it can be seen that the performance of the deep MOT method is mainly affected by two factors: object detection quality and re-ID feature learning, which correspond to the following two problems.

1) *Occlusion and Small Targets*: During the tracking process, the apparent features of partially occluded targets are severely lost, and small targets in the scene have insufficient features due to their small number of occupied pixels. The above issues have a significant impact on target detection and matching, thereby reducing the final performance of the tracking algorithm.
2) *The Inconsistent Target Features:* The features of the target are dynamically changing during tracking, and when the current and subsequent features change too much, there will be a problem of inconsistent features.

To solve the above problems, in addition to improving the accuracy of the detector, the work in recent years has mainly focused on re-ID Feature selection, in which motion and appearance are the most distinctive features. The most representative TBD methods include Bytetrack, OC-SORT, and so on, which mainly utilize motion feature as re-ID feature, while the most representative JDT methods include FairMOT, QDtrack, and so on, which mainly utilize appearance feature as re-ID feature. These methods will be compared with the proposed method as state-of-the-art in the subsequent experiments in Section IV.

### B. Video SAR Shadow Tracking

In recent years, shadow-based Video SAR moving target tracking has emerged one after another. Ding et al. [24] utilized the faster region-based convolutional neural network (Faster-RCNN) to detect shadows in each frame and filter these detections by density-based clustering to mitigate false alarms. After obtaining the detection box coordinates, it applied a bidirectional long-short-term memory network (Bi-LSTM) to predict the shadow trajectories offline. Zhou et al. [25] proposed the total variation loss-based smoothing LSTM to track the shadow in Video SAR. By applying the motion features, they have obtained considerable tracking results. However, this kind of method is designed for single shadow tracking.

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Although significant progress has been made in the field of optical video multiobject tracking, there is still great room for improvement in the performance of Video SAR multishadow tracking, especially in terms of MOT accuracy (MOTA) [26]. For multishadow tracking, Wang et al. [27] used optical MOT methods: FairMOT and CenterTrack for Video SAR tracking directly, but the tracking accuracy in real Video SAR data is low. Similar to the DeepSORT [28], Fang et al. [29] proposed multiscale Faster-RCNN to improve the detection accuracy and Siamese network to extract appearance features for data association. Yang et al. [19] and Bao et al. [20] improved the detector part in FairMOT and SiamMOT to enhance tracking accuracy, respectively. Yan et al. [30] used the YOLOv8 [31] detector and BOT-SORT [12] to achieve Video SAR multitarget instance segmentation and tracking.

One of the limitations of the above methods is that it does not incorporate feature interaction context when performing re-ID feature learning. Our proposed method combines the appearance and motion features of objects and allows them to interact with each other. By complementing multidimensional features, better association results can be generated and the number of accurately tracked objects and overall tracking accuracy can be improved.

### C. Graph Neural Network-Based Tracking

Given the demand for processing graph data in abundant practical applications, GNNs have emerged and are used to directly process graph structure data in an end-to-end manner, which have achieved significant success in many fields such as image or point cloud segmentation [32], [33], pose recognition [34], target tracking [35], pedestrian recognition [36], and motion prediction [37]. The graph is composed of nodes and edges, where nodes represent various data samples and edges represent the interrelationships between data samples. By utilizing GNN's node feature aggregation, node content information can be updated through interaction with neighboring nodes. For example, GCN [38] is a fundamental GNN model that extends convolutional operations from traditional data (such as images) to graph data. For a graph $G = \{V, E$, where $v$ represents nodes and $e$ represents edges, the core idea is to learn a function mapping $f(\cdot)$, through which the node $v_i$ in the mapping graph can aggregate its own feature $x_i$ with its neighbor feature $x_j$ to generate a new representation of node $v_i$.

Recently, GCN has been introduced in the computer vision field for optical video tracking. Ma et al. [39] used appearance features extracted by CNN as node features, and GCN was used to update node features, where nodes are individual detections per frame. After the GCN is updated, the cosine similarity of node features is used to calculate the adjacency matrix, and then, the detection is allocated to the known track in the way of maximum edge weight. Qi et al. [40] proposed a graph network to solve the problem of person re-ID, which is used to obtain discriminative features and ultimately determine the similarity between each pedestrian detection pair.

Although GNN has shown good performance in many fields, it has not yet been applied to Video SAR shadow tracking. To our knowledge, our work is the first attempt to use GNN for online Video SAR shadow tracking. The data association during the tracking process requires the detection results of previous and subsequent frames as nodes, the re-ID features corresponding to the detection results as node content, and the feature similarity between nodes as edge weights to construct a relationship graph and to match the nodes of the same target.

However, most existing works use unstable features for matching, which can easily result in inaccurate matching results. Different from prior work, our proposed method uses the graph structure and node content information as input and enables GNN to learn the node interaction function through end-to-end training, and trained GNN iteratively optimizes target features to make them more distinctive and achieve better matching results.

## III. Methodology

The deep MOT methods can be systematically categorized into three components: detection, feature extraction, and data association. Detection captures the current frame's results, feature extraction gathers attributes of new and tracked targets, and data association updates trajectories by aligning current detection with historical trajectories. This process forms the backbone of our MOT approach.

If given $M$ tracked objects $o_i \in O$ at previous frame $t-1$ where $i \in \{1, 2, \ldots, M$ and $N$ detected objects $d_j \in D$ at current frame $t$ where $j \in \{1, 2, \ldots, N$, our approach aims to improve feature extraction and data association to learn discriminative feature from $O$ and $D$ and then find the correct matching based on the pairwise feature similarity.

Our proposed framework is demonstrated in Fig. 2, which consists of motion feature extraction, appearance feature extraction, and GNN for data association. To achieve Video SAR multishadow tracking, we first apply the shadow detector to have detection bounding boxes for the current frame $t$ and past frames. The most noteworthy thing is that considering the rapid development and progress in the field of object detection [41], we did not restrict the detector when designing GNN-JFL. Instead, we used the detector as an optional plugin to facilitate the integration of more advanced detectors in the future. In this article, we take the Faster R-CNN [42] based on region of interest (ROI) Align and feature pyramid network (FPN) [43] as an example, and its specific details will be explained in Section IV-B. Then, the feature extraction component is applied to all objects in $O$ and $D$ to extract motion features and appearance features for next data association. Finally, the data association component takes the two kinds of features as input and constructs corresponding graphs with node being the object feature in the frame $t$ and $t-1$. After that, the GNN iteratively aggregates the node feature from the neighborhood and computes the affinity matrix for matching using edge regression.

### A. Joint Appearance and Motion Feature Extraction

To utilize the information for different modalities and learn discriminative feature, our proposed joint feature extractor

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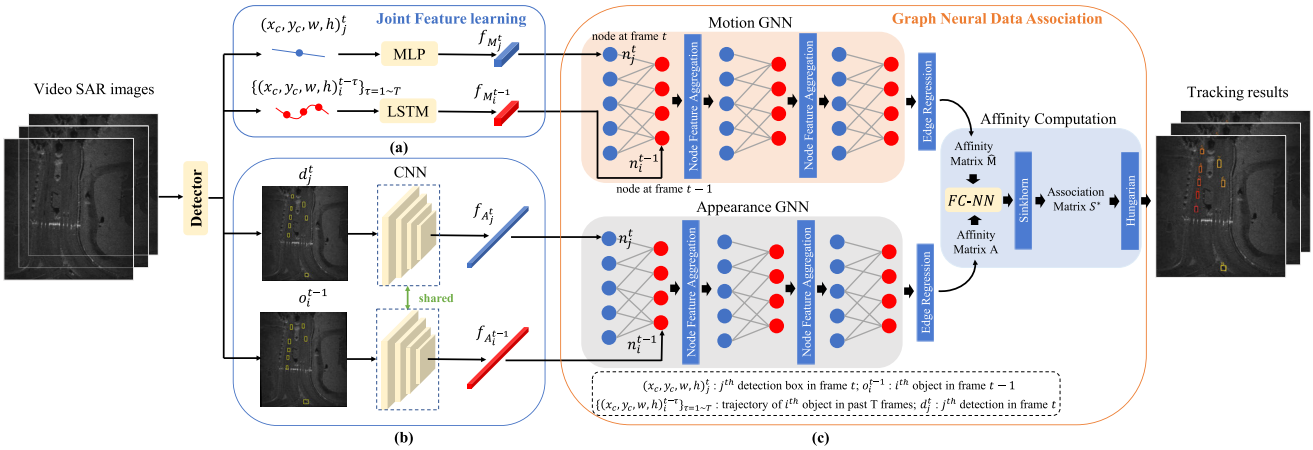



Fig. 2. Video SAR shadow tracking framework of GNN-JFL. (a) Motion feature extractor. (b) Appearance feature extractor. (c) GNN for data association. FC-NN is a fully connected network to compute the association matrix from the motion affinity matrix and appearance affinity matrix.

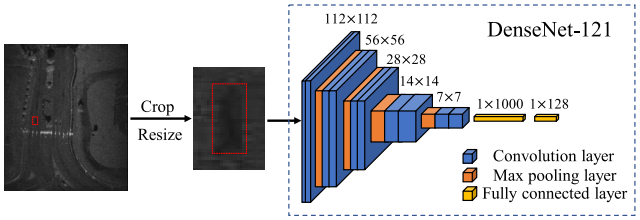



Fig. 3. Architecture of CNN in appearance feature extraction.

with two branches leverages both appearance and motion features, where one branch performs the appearance feature extraction and another branch performs the motion feature extraction.

*1) Appearance Feature Extraction:* As shown in Fig. 2(a), we use the bounding boxes detected by the detector as input for appearance feature extraction branch, and then, we get the image patch $d_j^t$ and $o_i^{t-1}$, which is cropped from the entire raw image based on the $j$th detection in frame $t$ and the $i$th detection in frame $t-1$, and finally, we feed them into a Siamese CNN (DenseNet [44]) architecture to extract the corresponding appearance feature $f_{A_i^{t-1}}$ and $f_{A_j^t}$. As shown in Fig. 3, we resize the cropped image patch for each object to $224 \times 224$ and use the DenseNet121 to extract the appearance feature with the dimensionality of 128.

*2) Motion Feature Extraction:* As shown in Fig. 2(b), we parameterize the detection as a tuple of $x_c$, $y_c$, $w$, $h$ where $(x_c, y_c)$ is the object center coordinate of the original image and $(w, h)$ denotes the width and height of the bounding boxes and uses it as input for motion feature extraction branch. Note that considering that the tracked object has associated trajectories in past frames, while the detected object does not, we used different motion feature extractors for the tracked and detected objects. For tracked object $o_i$, we apply the LSTM with two layers that take into the object's detections in past $T$ frames to obtain the feature. For detected object $d_j$, we use a two-layer multilayer perceptron (MLP) with the detection in frame $t$ as the input to extract features.

### B. Graph Neural Network for Data Association

Considering the differences in content and dimensions between the extracted appearance features and motion features, we adopt appearance GNN and motion GNN to process these two features. Also, the appearance GNN and motion GNN are both composed of three GAT layers with three attention heads.

*1) Graph Construction:* After feature extraction, we obtain $M$ features corresponding to tracked objects in frame $t-1$ and $N$ features corresponding to detected objects in frame $t$. These object features will serve as nodes in the constructed relationship graph. Then, we define the edges in the graph, which are the neighbors of the nodes. Due to that online MOT only performs data association between frame $t-1$ and $t$, matching only occurs between frames $t-1$ and $t$. Specifically, we only need to construct edges between nodes in different frames. In addition, for any tracked object $o_i$ in frame $t-1$, the possible matched detection $d_j$ in frame $t$ is most likely located in a nearby location. Therefore, we can set a maximum distance of $\text{Dist}_{\max}$ pixels and only construct edges when the distance between the detection centers of two nodes is less than the maximum distance. As a result, we construct a preliminary relationship graph for each consecutive frame as input to our graph matching network during training and testing.

*2) Node Feature Aggregation:* In order to preferably apply the feature interaction mechanism in GNN, we iteratively update the node feature by aggregating features from the neighborhood in every layer of the GNN, as shown in Fig. 2(c). The commonly used GCN is a method that combines topological structure and node attribute information in a graph to learn the embedded representation of nodes. However, GCN performs feature aggregation on each neighboring node equally when learning node embedding and cannot directly generalize to nodes that have not appeared during the training process, which may affect the discriminability of learned features. The GAT [45] used in this article has added a learnable attention weight to feature aggregation, allowing the network to focus its attention on neighborhood nodes with high similarity, which may be objects with the same ID. Compared to GCN, GAT is more suitable for MOT tasks, as it can exert more attention on effective targets in MOT. Its core idea is to generate the embedding vector of the target node by learning a function that aggregates the representations of neighboring

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nodes as follows:

$$n_v' = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{u\in N(v)} a_{uv}^{(k)} W^{(k)} n_u\right) \tag{1}$$

where $k = 1, \ldots, K$ is the number of heads in the GAT layer, $n_v'$ is the output feature of central node $v$, $W^{(k)}$ is the learnable parameter, and $a_{uv}^{(k)}$ is the attention score. $\sigma$ is the Leaky ReLU activation function, $\mathrm{N}(v)$ represents the node $v$'s neighbors, and $n_u$ is the input feature of neighbor node $u$. GAT adopts the multihead attention mechanism to update node features, and the attention score of each head is calculated by the following equations:

$$e_{uv}^{(k)} = \mathrm{LeakyReLU}\left(\vec{a}_k^{\,T}\left[W^{(k)} n_u || W^{(k)} n_u\right]\right) \tag{2}$$

$$a_{uv}^{(k)} = \frac{\exp\left(e_{uv}^{(k)}\right)}{\sum_{z\in \mathrm{N}(v)} \exp\left(e_{zv}^{(k)}\right)} \tag{3}$$

where $\vec{a}_k$ is the learnable parameter as $W^{(k)}$ and $||$ represents the feature concatenation operation.

*3) Edge Regression:* After aggregating GNN node features, we obtain the optimized relationship graph. Edge regression is used to compute the $M \times N$ appearance affinity matrix $A$ and motion affinity matrix $\hat{M}$ based on the pairwise similarity of the features extracted from $M$ tracked objects in frame $t-1$ and $N$ detected objects in frame $t$. We employ the cosine similarity metric for measuring the feature similarity, which is a conventional metric used in the MOT community. It takes the two node features as input and outputs a scalar value between 0 and 1 as the pairwise similarity score

$$A_{ij} = \frac{n_i^{t-1} \cdot n_j^t}{\left|n_i^{t-1}\right|\left|n_j^t\right|} \tag{4}$$

where $n_i^{t-1}$ and $n_j^t$ are the $i$th node features ($i \in \{1, 2, \ldots, M\}$) in frame $t-1$ and the $j$th node features ($j \in \{1, 2, \ldots, N\}$) in frame $t$, and $\cdot$ represents the vector dot product.

*4) Affinity Computation:* As shown in Fig. 2(c), we use the appearance affinity matrix $A$ extracted by appearance GNN along with the motion affinity matrix $\hat{M}$ extracted by motion GNN to compute the final affinity of a tracklet $o_i$ to be associated with a detection $d_j$ through the affinity computation module. This is different from the data association part of existing Video SAR shadow MOT methods. Prevailing Video SAR shadow tracking approaches predominantly rely on the appearance features of tracked objects during the similarity computation phase and employ Kalman filtering in the subsequent tracking process. Consequently, such methods incompletely utilize the information available in the motion features of tracked objects. In this module, given the equal importance of the motion features and appearance features of $o_i$ and $d_j$ in determining their association affinity, we feed the appearance cosine similarity score $A$ and the motion cosine similarity score $\hat{M}$ to a fully connected neural network (FC-NN), which produces the intermediate affinity matrix $S$. The FC-NN uses a fully connected layer and a nonlinear activation function ReLU.

Each entry within the affinity matrix $S$ serves as the quantitative measure characterizing the suitability of assigning tracklet $m$ to detection $n$. Notably, certain detections may not find an association with any tracklet, while conversely, some tracklets may fail to establish a meaningful connection with any detection. These unassociated scenarios are indicative of object "birth" and "death" within the context of MOT. To account for these critical event states, we augment the matrix $S$ by appending a supplementary row vector and column vector, primarily composed of binary elements (0 and 1). Specifically, a value of "1" within these appended vectors serves to symbolize the occurrence of object "birth" or "death." Also, the optimal $S$ is subject to the following MOT constraints:

$$\sum_{n=1}^{N+1} s_{m,n} = 1 \quad \forall m \in [1, \ldots, M] \tag{5}$$

$$\sum_{m=1}^{M+1} s_{m,n} = 1 \quad \forall n \in [1, \ldots, N]. \tag{6}$$

The Sinkhorn algorithm is a classic algorithm for solving the optimal transmission scheme problem, which is used for optimal data transmission between two different distributions. To enforce the unique MOT constraints of the bipartite graph matching problem mentioned above, our proposed method utilizes the Sinkhorn algorithm [46] in the affinity computation module to map the unconstrained affinity distribution learned by GNN to the affinity distribution that conforms to MOT constraints. The affinity distribution learned by GNN is represented by the affinity matrix $S$. Also, the Sinkhorn algorithm is leveraged to iteratively normalize the affinity matrix $S$ to satisfy the MOT constraints (5) and (6). In each iteration, matrix $S$ is transformed row by row using (7) and column by column using (8)

$$s_{m,n}' = \lambda_m \frac{e^{l*s_{m,n}}}{\sum_{k=1}^{N+1} e^{l*s_{m,k}}}, \lambda_m = \begin{cases} 1, & m \in [1, \ldots, M] \\ N, & m = M+1 \end{cases} \tag{7}$$

$$s_{m,n}' = \mu_m \frac{e^{l*s_{m,n}}}{\sum_{k=1}^{M+1} e^{l*s_{k,n}}}, \mu_m = \begin{cases} 1, & n \in [1, \ldots, N] \\ M, & n = N+1 \end{cases} \tag{8}$$

where $l$ is a hyperparameter. After a fixed number of iterations is completed, the supplemented row and column need to be removed to obtain the final association matrix $S^*$ that is consistent with the size of matrix $S$. Since the Sinkhorn algorithm is completely differentiable, $S^*$ can be directly fed to the loss function of our approach, allowing the network to learn the optimal matching.

During GNN model training, the normalized affinity matrix $S^*$ can be applied in the affinity loss function for end-to-end learning of the relationships between objects. During GNN model testing, the normalized affinity matrix $S^*$ can be applied as the cost matrix for the Hungarian algorithm to predict object associations.

### C. Loss Function

Our proposed network employs the following two losses during training: 1) batch triple loss $L_{\mathrm{tri}}$ and 2) affinity loss $L_{\mathrm{aff}}$. We can summarize the entire loss function $L$ as follows:

$$L = L_{\mathrm{tri}} + L_{\mathrm{aff}}. \tag{9}$$

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**Algorithm 1** Multishadow Tracking Framework of GNN-JFL for Video SAR

**Input:** A video sequence $s$ with $N$ frames;
**Output:** Tracking results $\chi$ of video SAR;
**Hyperparameters Initialization:** $Dist_{max} = 3$, $\mu_1 = 0.5$, $\mu_2 = 0.1$, $T = 5$, $a = 5$;
Tracking parameters Initialization: $\chi = \{\}$, $\mathrm{B} = \{\}$, $\Lambda = \{\}$;

/* initialize tracklets based on the first frame */
Feed the first frame into the detector to obtain the detection result $\{Det\}_{t=1}$ with a confidence threshold $\mu_1$;
Feed the detection box into MLP to obtain the corresponding motion feature $\{f_M\}_{t=1}$;
Feed the cropped image into CNN to obtain the corresponding appearance feature $\{f_A\}_{t=1}$;
/* Tracking Loop */
**for** frame $t$ in $s$ **do**
  Feed frame $t$ into the detector to obtain the detection result $\{Det\}_t$ with a confidence threshold $\mu_1$;
  Feed the detection box into MLP to obtain the corresponding motion feature $\{f_M\}_t$;
  Feed the cropped image into CNN to obtain the corresponding appearance feature $\{f_A\}_t$;
  /* tracklets matching */
  Construct the motion graph with $\{Det\}_{t-1}$, $\{Det\}_t$ as nodes and $\{f_M\}_{t-1}$, $\{f_M\}_t$ as node features;
  Construct the appearance graph with $\{Det\}_{t-1}$, $\{Det\}_t$ as nodes and $\{f_A\}_{t-1}$, $\{f_A\}_t$ as node features;
  Feed the graph into GNN to obtain the motion affinity matrix $\hat{M}$ and appearance affinity matrix $A$;
  Acquire the association matrix $D$ through FC-NN and Sinkorn algorithm;
  Match by using Hungarian algorithm based on $D$ with a matching threshold $\mu_2$;
  **if** matched **then**
    update $\chi$
  **else**
    update the unmatched tracklets $\Lambda$ by unmatched detections in frame $t-1$;
    update the new birth tracklets B by unmatched detections in frame $t$;
  /* preparation for the next loop */
  **if** $t \geq T$ **then**
    update $\{f_M\}_t$ with the motion feature of historical trajectory obtained by LSTM;
  /* Tracking Management */
  **if** find the match for object $o$ in $\Lambda$ within $a$ frames **then**
    remove $o$ from $\Lambda$
  **if** find the match for object $p$ in $B$ within continuous $a$ frames **then**
    remove $p$ from $B$
**return** result $\chi$

*1) Batch Triplet Loss:* In order to learn more discriminative appearance and motion features for matching, we first apply a batch triplet loss to extracted features from joint feature learning. For feature $f_i^{t-1}$ that has a matched feature $f_j^t$, the batch triplet loss is defined as

$$L_{\text{tri}} = \max\left(\left\|f_i^{t-1} - f_j^t\right\| - \left\|f_i^{t-1} - f_s^t\right\| - \left\|f_r^{t-1} - f_j^t\right\| + \alpha, 0\right) \tag{10}$$

where $\alpha$ is the margin of the triplet loss and $f_s^t$ and $f_r^{t-1}$ are the features with different IDs from $f_j^t$ and $f_i^{t-1}$ in frame $t$ and $t-1$, respectively. $f_i^{t-1}$ is the motion or appearance feature, where $i$ is the object ID and $t-1$ is the video frame. Different from the original definition in [47], the above batch triplet loss only has one positive term $\|f_i^{t-1} - f_j^t\|$ that makes features with the same ID to be similar and has two negative terms that force the features with different IDs to be different. For the case where $f_i^{t-1}$ does not have a matched feature, the first term in (10) is removed and only the remaining two negative terms in the loss $L_{\text{tri}}$ are minimized. The final loss is calculated by adding the batch triplet loss of motion features and appearance features.

*2) Affinity Loss:* Besides applying the batch triplet loss to supervise the joint feature extraction module, we also employ an affinity loss $L_{\text{aff}}$ to directly supervise the final output $S^*$ of the network. We construct the ground-truth affinity matrix $O$ before training, where $o_{mn}$ is one of the elements. $o_{mn}$ is assigned a value of 1 only when tracklet $m$ and detection $n$ have the same id, and $o_{mn}$ is assigned a value of 0 in other cases. As the ground-truth affinity matrix $O$ only has integer 0 or 1 on all the entries, the prediction of the affinity matrix can formulate as a binary classification problem. The affinity loss adopts the weighted binary cross-entropy loss of each element in the affinity matrix $S^*$ as shown in the following:

$$L_{\text{aff}} = \frac{-1}{MN}\sum_{m}^{M}\sum_{n}^{N} w * s'_{m,n}\log o_{m,n} + \left(1 - s'_{m,n}\right)\log\left(1 - o_{m,n}\right) \tag{11}$$

where $w$ is a weight hyperparameter to balance the imbalance among 1's and 0's in the ground-truth labels $O$.

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### D. Tracking Process

After training GNN-JFL with paired SAR images and continuously optimizing losses, the trained GNN-JFL can be used to test new Video SAR images. During the test stage, GNN-JFL mainly completes tracking through the process shown in Algorithm 1, and the specific steps are as follows.

*Step 1:* Input Video SAR images.

*Step 2 (Detect Shadows):* Use a detector (Faster-RCNN) to perform shadow detection on each Video SAR image and obtain the detection result with a confidence threshold greater than $\mu_1$.

*Step 3 (Extract Features):* After obtaining the detection boxes of each frame, GNN-JFL applies a joint feature extraction network with two branches. The appearance feature extraction branch inputs the shadows in the detection boxes of frame $t-1$ and frame $t$ into CNN (DenseNet) to obtain their appearance features. The motion feature extraction branch inputs the historical trajectory in frame $t-1$ and the detection box position information $(x, y, w, h)$ in frame $t$ into LSTM and MLP, respectively, to obtain the corresponding motion features. By utilizing complementary information from different modalities, joint feature extraction networks can learn better feature representations, which is beneficial for improving subsequent tracking accuracy.

*Step 4 (GNN for Features Interaction):* Use the extracted appearance and motion features to construct feature relationship graphs separately, where nodes represent the detection results in frames $t-1$ and $t$, node features represent the corresponding features of the detection results, and edges represent the similarity between node features. Then, input the feature relationship graph into the trained appearance GNN and motion GNN to complete the feature interaction.

*Step 5 (Data Association):* By calculating the cosine similarity of node features in the optimized feature relationship graph, the appearance affinity matrix $A$ and the motion affinity matrix $M$ are obtained. Subsequently, the fused affinity matrix $S$ was obtained using FC-NN. To satisfy the MOT constraint, we used the Sinkhorn algorithm to regularize $S$ and got the final affinity matrix $D$. Finally, the Hungarian algorithm is used to match the detection in frame $t-1$ and frame $t$ based on $D$, and the ID is assigned to the matching result. Our method introduced the feature aggregation mechanism of GNN to optimize the feature relationship graph and achieve better matching.

*Step 6 (Tracking Management):* Although JFL and GNN feature interaction can help solve confusion and omissions in matching, the presence of false targets and missed detections in the detection results caused by detector errors may still affect data association, resulting in matching the tracking object to false targets or the inability to find a match due to missed detections. To avoid the above situation, we have added the tracking management step to reduce false positives (FPs) and false negatives (FNs). Specifically, we maintain two sets, namely, unmatched trajectory $\Lambda$ and new born trajectory $B$, consisting of unmatched detections in frames $t-1$ and $t$, respectively. If the object in $\Lambda$ cannot find a match within $a$ frames, we assume that the trajectory has terminated. Otherwise, we assume that the trajectory still exists and removes it from $\Lambda$ and assign an ID. If the object in $B$ can find a match within continuous $a$ frames, we consider the trajectory to be a new born trajectory, remove it from $B$, and assign an ID. Otherwise, we consider the trajectory to be a false trajectory.

## IV. Experiments

### A. Settings

*1) Dataset:* Our first experimental data come from Sandia National Laboratories [48], which is a Video SAR footage of a highway where the moving target is a vehicle traveling on the road. The SNL data consist of 900 consecutive SAR images, with a video frame rate of 30 FPS. The SNL data are collected by a Video SAR, which operates in the Ku-band with a center frequency of 16.7 GHz. The image resolution is 0.1 m and the image size is $660 \times 720$. We divide the above data into nine subsequences, each subsequence having 100 consecutive frames, and solve the problem of scene changes in the video through image registration. Six subsequences are used as training sets, and the other three are used as test sets. Our second experimental data come from China Aerospace Science and Industry Corporation 23 research institute. These data were obtained from the spotlight SAR on an aircraft flying at a height of 3000 m. Similar to SNL data, this is a Video SAR shot of a circular road, where the moving target is also the vehicle traveling on the road. The carrier frequency and velocity of aircraft are Ka-band and 80 m/s. These CASIC data consist of 369 SAR images. The image resolution is 0.15 m and the image size is $1000 \times 1000$ pixels. The complete data are divided into four subsequences, with three used as the training set and one used as the testing set. Significantly, our method adopts Video SAR image sequences for tracking, with no restrictions on SAR imaging modes and imaging scenes.

*2) Evaluation Metrics:* Seven commonly used evaluation indicators in the MOT field are used to verify the tracking performance of GNN-JFL, including MOTA, identity switches (IDSWs), FPs, FNs, identification precision (IDP), identification recall (IDR), identification F1 score (IDF1), and frame per speed (FPS) [26]. MOTA is the most important indicator in MOT evaluation that measures all tracking errors caused by the tracker over all frames, including FN, FP, and IDSW. FN calculated the number of targets that were not tracked in ground truth. FP calculated the number of false tracking targets in the tracking results. IDSW calculated the number of times ID switching occurred in the tracking results. MOTA can be expressed as

$$\mathrm{MOTA} = 1 - \frac{\sum_t (\mathrm{FN}_t + \mathrm{FP}_t + \mathrm{IDSW}_t)}{\sum_t \mathrm{GT}_t} \tag{12}$$

where $\mathrm{FN}_t$, $\mathrm{FP}_t$, $\mathrm{IDSW}_t$, and $\mathrm{GT}_t$ represent the number of FNs, FPs, IDSWs, and ground truth at frame $t$, respectively. MOTA will become larger if the tracking performance is better. FPS is the average running time of the tracking algorithm for one frame. To assess the accuracy of the target identity during the tracking process, we employ the IDP, IDR, and IDF1 metrics for evaluation. Concretely, IDP (IDR) is defined

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as the proportion of correctly identified detections among the computed (or ground truth) detections. IDF1 is computed as the ratio of correctly identified detections over the average number of ground truth and computed detections [26] and is formulated as follows:

$$\mathrm{IDP} = \frac{\mathrm{IDTP}}{\mathrm{IDTP} + \mathrm{IDFP}} \tag{13}$$

$$\mathrm{IDR} = \frac{\mathrm{IDTP}}{\mathrm{IDTP} + \mathrm{IDFN}} \tag{14}$$

$$\mathrm{IDF1} = \frac{2\mathrm{IDTP}}{2\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}} \tag{15}$$

where IDTP, IDFP, and IDFN represent the total number of identification true positives, FPs, and FNs, respectively.

### B. Implementation Details

*1) Object Detection:* Due to the impact of detection results on tracking performance, in order to better validate the effectiveness of the proposed method, we chose the commonly used Faster R-CNN as the detector. Faster R-CNN extracts image features through the backbone network, uses FPN to obtain the top-down feature pyramid, and then classifies and regresses the obtained detection box. Its backbone network adopts the ResNet-50 structure [49], and the channel number of feature maps generated by the FPN is 256/512/1024/2048. The network was optimized with the following loss function:

$$L_{\mathrm{det}} = L_{\mathrm{rpn}} + \lambda_1 L_{\mathrm{cls}} + \lambda_2 L_{\mathrm{reg}} \tag{16}$$

where the RPN loss $L_{\mathrm{rpn}}$, classification loss $L_{\mathrm{cls}}$, and regression loss $L_{\mathrm{reg}}$ remain the same as the original article and the loss weights $\lambda_1$ and $\lambda_2$ are set to 1.0 by default. The score threshold $\mu_1$ of the detector during the testing phase is set to 0.5.

*2) Joint Feature Extractor:* For the appearance branch, we resize the cropped image patch for each object to $224 \times 224$ and use the DenseNet121 to extract the appearance feature with the dimensionality of 128. For the motion branch, we use a two-layer LSTM with a hidden size of 16 and a number of past frames $T = 5$ for tracked objects, which considers both computational cost and performance. The correlation between the historical trajectory within $T = 5$ frames and the current detection is strong and the computational cost is low. For tracked objects that only have associated detections in past $R(< T)$ frames, we repeat the earliest detection $T - R$ times so that the objects can have $T$ frames of detections. For detected objects, we employ a two-layer MLP (feature size 4 $\Rightarrow$8$\Rightarrow$16 in motion branch).

*3) Graph Neural Network and Affinity Computation:* We use $\mathrm{Dist}_{\max} = 30$ in our sparse graph construction. We use three GAT layers with each layer having node feature with a dimensionality of $128 \Rightarrow 256 \Rightarrow 128$ for the appearance branch and $16 \Rightarrow 32 \Rightarrow 16$ for the motion branch. For edge regression, we use a two-layer MLP with hidden feature dimension of $128 \Rightarrow 32 \Rightarrow 1$ for the appearance branch and $16 \Rightarrow 4 \Rightarrow 1$ for the motion branch. For batch triplet loss, we use the margin $\alpha = 10$ by default. For the tracking management, we use $\Lambda = B = 5$ with the same setting reason as $T$. When no shadows are tracked in consecutive $\Lambda$ frames, the trajectory is considered terminated. When unmatched shadows are tracked in consecutive $B$ frames, it is considered that the trajectory is born. The matching threshold $\mu_2$ of Hungarian algorithm is set to 0.1.

TABLE I
EXPERIMENTAL PARAMETERS

| Hyper-parameters | Detector | Tracker |
|---|---|---|
| Learning rate | 0.008 | 0.001 |
| Weight decay | 0.0001 | 0.0001 |
| Momentum | 0.9 | - |
| Epochs | 12 | 10 |
| Batch size | 2 | 4 |
| Optimizer | SGD | Adam |

*4) Training Setup:* All the experiments in this article were implemented on an Intel i9-9900K CPU and an NVIDIA GTX3070Ti GPU. The software experimental environment was Windows 10 and Pytorch 1.8 with python 3.8. The proposed method was trained in two stages, with the parameters shown in Table I. The first stage trains the detector, and after the detector is trained, the second stage trains the tracker (joint feature extractor and graph neural network). When training the detector, we use the ImageNet pretraining weight [50] of ResNet-50 to initialize the backbone network. The stochastic gradient descent (SGD) [51] optimizer is used to optimize the detection loss function. When training the tracker, the pretraining weights of DenseNet-121 are used to initialize the appearance feature extraction CNN. The adaptive moment estimation (Adam) optimizer is used to optimize the loss function in Section III-C.

### C. Comparisons With Other Methods in SNL Dataset

Several state-of-the-art methods have been compared with GNN-JFL to verify its effectiveness, including Bytetrack [11], BOT-SORT [12], OC-SORT [13], SiamMOT [14], FairMOT [15], and QDtrack [16]. We adopted FairMOT and SiamMOT with basic detectors to fairly compare the performance of the tracker part in [19] and [20].

In order to qualitatively analyze the tracking results of various methods, the shadow tracking results of comparative methods and GNN-JFL are shown in Fig. 4. In this experiment, we used sequence 1 from the test set, which consists of 100 frames and 12 target trajectories, with a maximum number of targets per frame of 9. The number in the top-left corner of the image represents its frame order in SNL Video SAR data, namely, 40, 45, 50, 60, and 65. The red dashed circle indicates missed detection, the red dashed rectangle indicates false alarm, and the red arrow indicates ID switching. We mark shadow targets with different IDs by drawing rectangular boxes of different colors on the image. The numbers on the rectangular boxes are generated by the network to represent the ID of the shadow, and the same numbers represent the same shadow.

From Fig. 4, we can draw the following conclusion.

1) As mentioned in Section III-D, unassociated and low-confidence detections during the tracking process will

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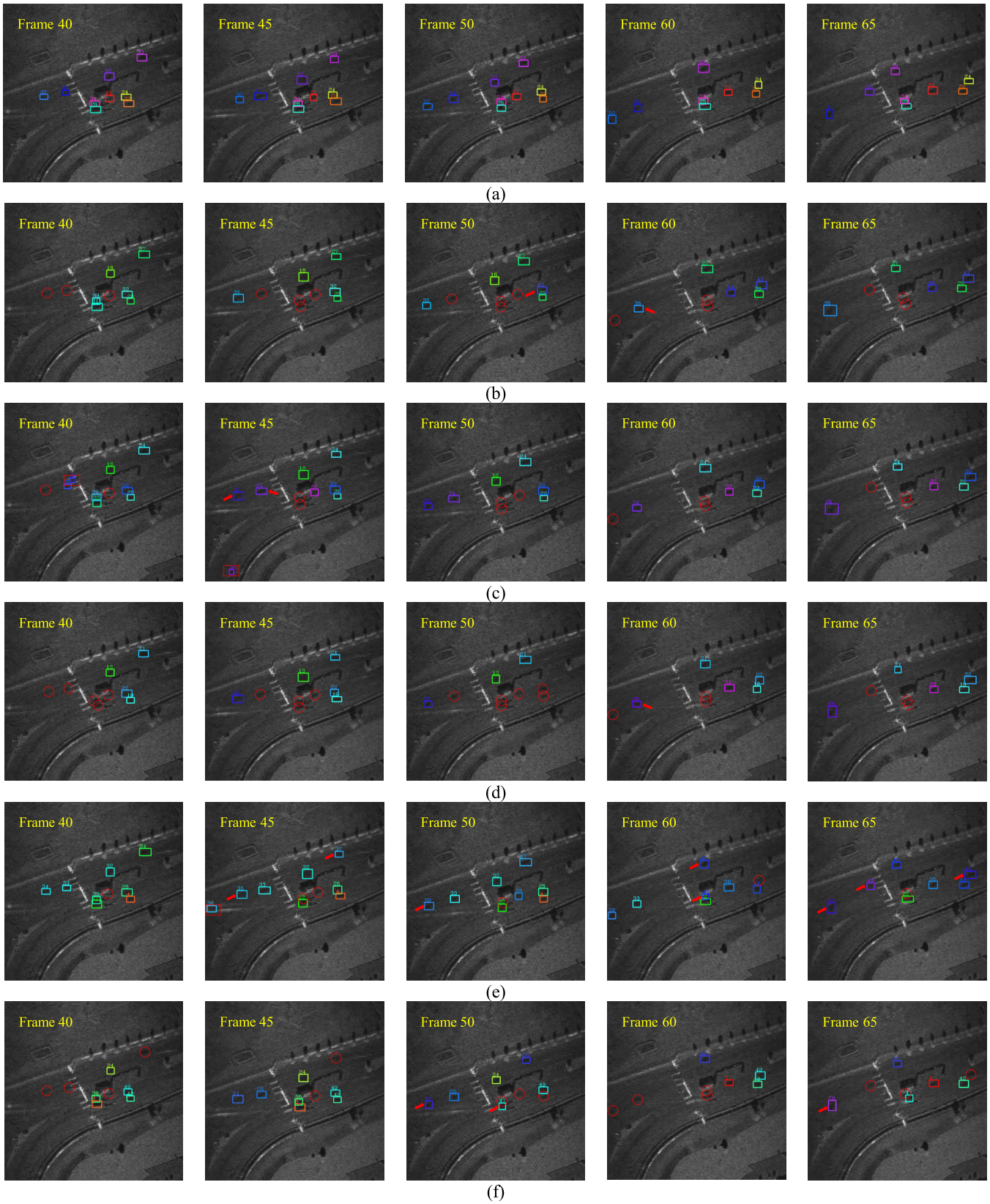



Fig. 4. Tracking results with different methods in SNL data. (a) Ground truth. (b) Bytetrack. (c) BOT-SORT. (d) OC-SORT. (e) SiamMOT. (f) FairMOT. (g) QDtrack. (h) GNN-JFL (ours). The red dashed circle indicates missed detection, the red dashed rectangle indicates false alarm, and the red arrow indicates ID switching.

not be given an ID, resulting in missed detections. GNN-JFL can avoid massive detection losses caused by matching errors. For example, ByteTrack, OC-SORT, and FairMOT all have many missed detections in frames

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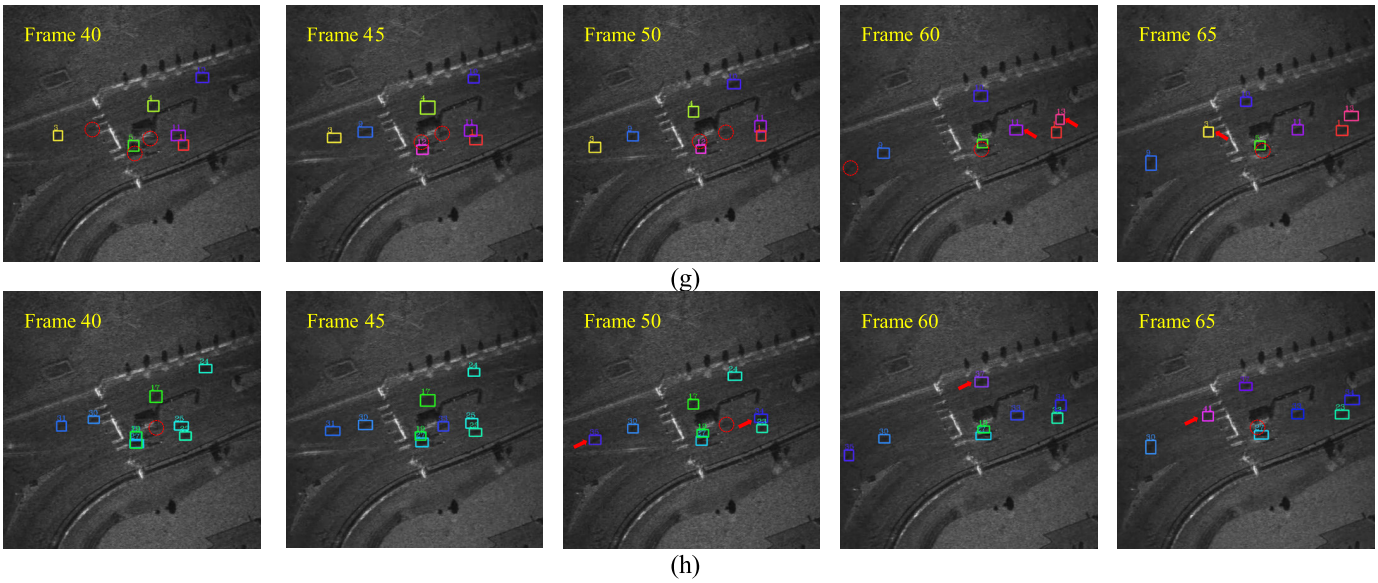



Fig. 4. *(Continued.)* Tracking results with different methods in SNL data. (a) Ground truth. (b) Bytetrack. (c) BOT-SORT. (d) OC-SORT. (e) SiamMOT. (f) FairMOT. (g) QDtrack. (h) GNN-JFL (ours). The red dashed circle indicates missed detection, the red dashed rectangle indicates false alarm, and the red arrow indicates ID switching.

TABLE II
SHADOW TRACKING PERFORMANCE OF DIFFERENT METHODS IN SNL DATA

| Tracker | MOTA(%)↑ | IDSW↓ | FP↓ | FN↓ | IDF1(%)↑ |
|---|---|---|---|---|---|
| SiamMOT | 44.5 | 63 | 349 | 529 | 45.0 |
| FairMOT | 46.6 | 143 | 134 | 628 | 48.2 |
| ByteTrack | 47.3 | 39 | 87 | 767 | 52.0 |
| QDTrack | 52.3 | 35 | 251 | 522 | 54.2 |
| BOT-SORT | 55.7 | 77 | 284 | 389 | 58.2 |
| OC-SORT | 56.5 | **18** | **58** | 661 | 60.8 |
| Ours | **60.7** | 43 | 237 | **385** | **63.6** |

40, 45, 50, 60, and 65, while GNN-JFL can correctly associate all detections and assign IDs to track all shadows.

2) Compared to GNN-JFL, ByteTrack, SiamMOT, and FairMOT have more ID switches, which typically occur in situations with background occlusion and changes in shadow appearance. This reduces tracking accuracy and trajectory continuity.

3) For some difficult-to-detect shadow targets, such as the shadow with id 34 in frame 65, their low distinguishability from the background results in tracking errors caused by detector miss in these methods, which requires further research. Therefore, in summary, we believe that GNN-JFL has the best tracking performance.

To compare tracking performance more intuitively, we further present the final tracking trajectories of these methods in all test sets in Fig. 5. Fig. 5(a) shows the real ground trajectory of the test set, and we can see that the shadow disappears for a long time due to occlusion when the target passes through the central highlighted area. In addition, there are also shadows caused by occlusion in other areas that disappear in the short term, and due to poor video quality, shadows can also be submerged by the background and disappear. Therefore, some trajectory curves are not continuous.

From Fig. 5, the following conclusions can be drawn.

1) Compared with other methods, GNN-JFL has the least missed detections and has achieved tracking of most shadow targets in the video, while Bytetrack, OC-SORT, and FairMOT methods have significantly large missed detections.

2) Compared with other methods, the ID switching of GNN-JFL is relatively low, and FairMOT has the most ID switching. Increasing the IDSW value will not significantly reduce MOTA, but in practical applications, frequent ID switching is unacceptable.

3) Compared with other methods, the tracking trajectory of GNN-JFL is more continuous and the IDF1 of GNN-JFL is the highest, indicating that the proposed method has more accurate data association and fewer matching errors.

4) The tracking performance of GNN-JFL is superior to other methods. First, joint motion-appearance feature learning is proposed to enhance the discriminability of shadow features for subsequent feature interactions. Second, motion GNN and appearance GNN are proposed to optimize the extracted shadow features to achieve more accurate data association. Finally, using GAT to achieve feature aggregation by

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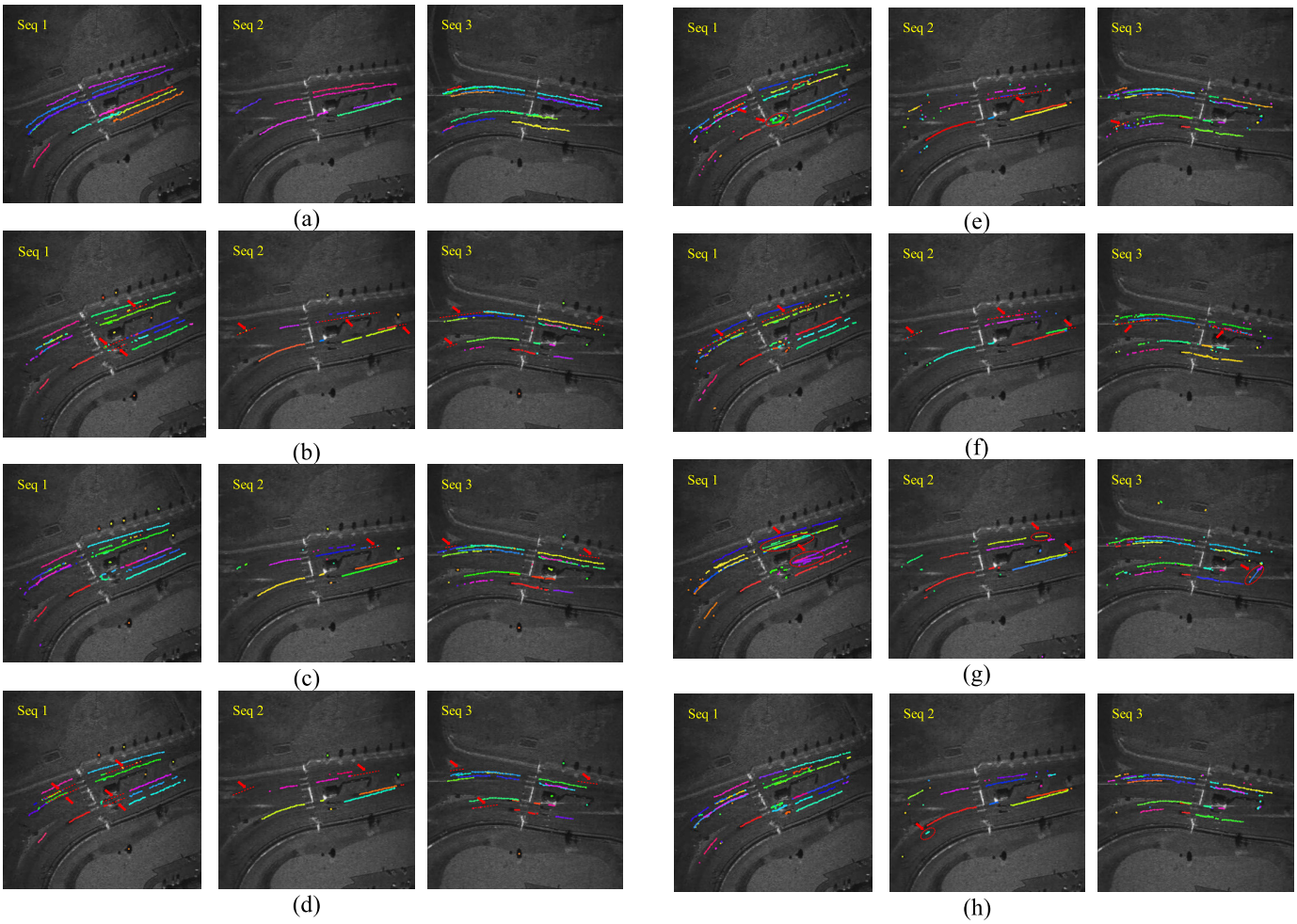



Fig. 5. Tracking trajectory with different methods in SNL data. (a) Ground truth. (b) Bytetrack. (c) BOT-SORT. (d) OC-SORT. (e) SiamMOT. (f) FairMOT. (g) QDtrack. (h) GNN-JFL (ours). The red dashed elliptic indicates a false trajectory, and the red dashed line indicates a missed trajectory.

focusing on neighboring nodes of interest, further suppressing mismatches. The effectiveness of the above improvements is verified by the best tracking performance of GNN-JFL.

In order to quantitatively analyze the tracking results of each method, we evaluated these methods through the evaluation indicators mentioned in Section IV-A. Table II shows the comparison results of tracking performance, and the tracking results of the entire test set were utilized for indicator calculation. All experiments were conducted under the same hardware and software conditions. MOTA is the most important indicator for evaluating the performance of MOT. As shown in Table II, GNN-JFL outperforms other networks in MOTA, with MOTAs 16.2%, 14.1%, 13.4%, 8.4%, 5%, and 4.2% higher than SiamMOT, FairMOT, ByteTrack, QDTrack, BOT-SORT, and OC-SORT, respectively.

QDTrack uses quasi-dense similarity learning for feature interaction after appearance feature extraction when building a tracker and improves the discriminability of features through comparative learning. However, QDTrack only utilizes the appearance features of SAR images for data association, which inevitably results in matching errors and detection loss due to changes in shadow appearance. In contrast, although GNN-JFL and QDTrack have the same detector (Fast-RCNN), GNN-JFL uses joint learning of motion and appearance features for data association, resulting in higher MOTA. FairMOT applies a single-feature extraction network to simultaneously achieve target detection and re-ID tasks. Although it has high computational efficiency, sharing the feature extraction module can lead to inaccurate extraction of shadow appearance features, resulting in the most severe ID switching. SiamMOT builds the tracker based on a Siamese network of the backbone network and applies the correlation operation to calculate the shadows' appearance feature similarity for data association. SiamMOT's incorrect tracking results are attributed to its failure to constrain the extracted appearance features in the loss function during the training phase.

Owing to the Kalman filter (KF) having accumulative noise, BoT-SORT will accumulate tracking errors when processing sequence datasets. For this reason, OC-SORT improves KF for data association, which fixes the accumulative noise during occlusion. However, during the association process, the detection that does not meet the prediction of KF will not be given an ID. Therefore, the ID accuracy of their association results is high, and FP and IDSW are the lowest, but the missing detection is relatively high among these methods. On the contrary, GNN-JFL uses GNN to achieve optimal matching during the association process, thus minimizing FN. In addition, due to the fact that low-confidence detection boxes in SAR image shadow detection

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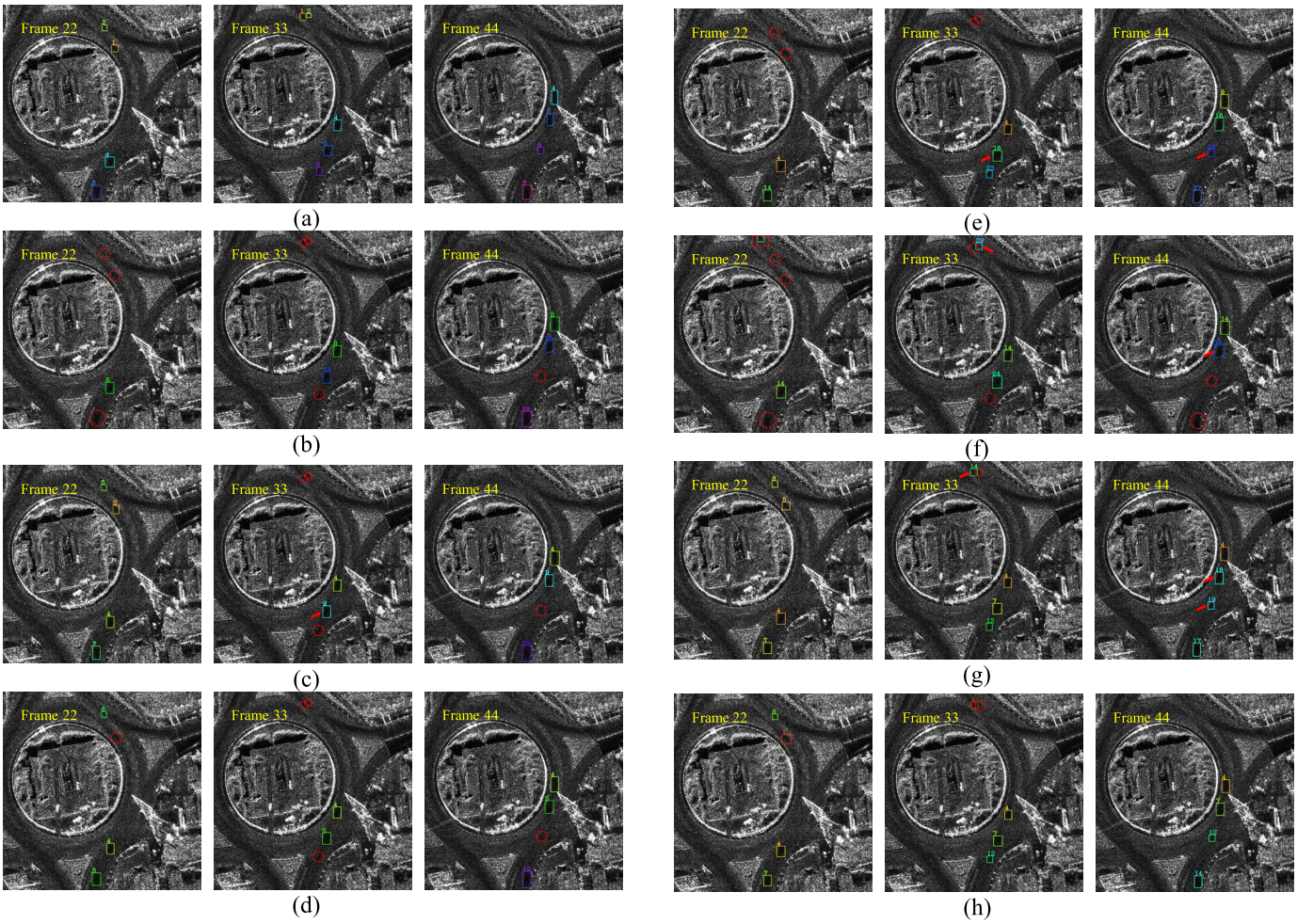



Fig. 6. Tracking results with different methods in CASIC data. (a) Ground truth. (b) Bytetrack. (c) BOT-SORT. (d) OC-SORT. (e) SiamMOT. (f) FairMOT. (g) QDtrack. (h) GNN-JFL (ours). The red dashed circle indicates missed detection, the red dashed rectangle indicates false alarm, and the red arrow indicates ID switching.

are often false targets, ByteTrack not only performs data association on high-confidence detection boxes but also on low-confidence detection boxes, which may result in poor tracking performance.

Although the fastest tracker currently is FairMOT, with a tracking speed of 21 frames/s, GNN-JFL only achieves 10 frames/s. However, GNN-JFL can achieve the most accurate MOT of SNL Video SAR data at the expense of some inference speed. We believe that sacrificing speed slightly to achieve significant MOTA improvement is worthwhile.

### *D. Comparisons With Other Methods in CASIC 23 Datasets*

To further verify the tracking performance and robustness of GNN-JFL, we conduct experiments on another real Video SAR dataset.

The qualitative and quantitative shadow tracking results of the six comparative methods and our GNN-JFL are shown in Fig. 6 and Table III, respectively. In addition, due to a large amount of overlapping trajectories, we further present the 3-D tracking trajectories of these methods in Fig. 7 to compare their tracking performance.

In this experiment, we used the test sequence from the CASIC 23 dataset, which consists of 101 frames and seven target trajectories. From Fig. 6, GNN-JFL correctly tracks all targets except for the target with ID 8 and has the least missed detections and ID switching compared to other methods. From Fig. 7, GNN-JFL successfully tracks all seven trajectories, while other methods miss at least one trajectory and even produce false trajectories.

CASIC 23 data have lower resolution and stronger background interference than SNL data, so the tracking accuracy of all methods in Table III has decreased. Furthermore, the low discrimination between shadows and background results in the

TABLE III
SHADOW TRACKING PERFORMANCE OF DIFFERENT METHODS IN CASIC DATA

| Tracker | MOTA(%) | IDSW | FP | FN | IDF1(%) |
|---|---|---|---|---|---|
| SiamMOT | 33.4 | 57 | 17 | 185 | 41.3 |
| FairMOT | 31.6 | 24 | 24 | 218 | 29.5 |
| ByteTrack | 35.0 | 7 | **11** | 235 | 33.6 |
| QDTrack | 41.4 | 20 | 42 | 166 | 33.3 |
| BOT-SORT | 46.5 | 9 | 28 | 171 | 55.7 |
| OC-SORT | 44.0 | **4** | 13 | 201 | 44.4 |
| Ours | **50.1** | 7 | 44 | **143** | **64.8** |

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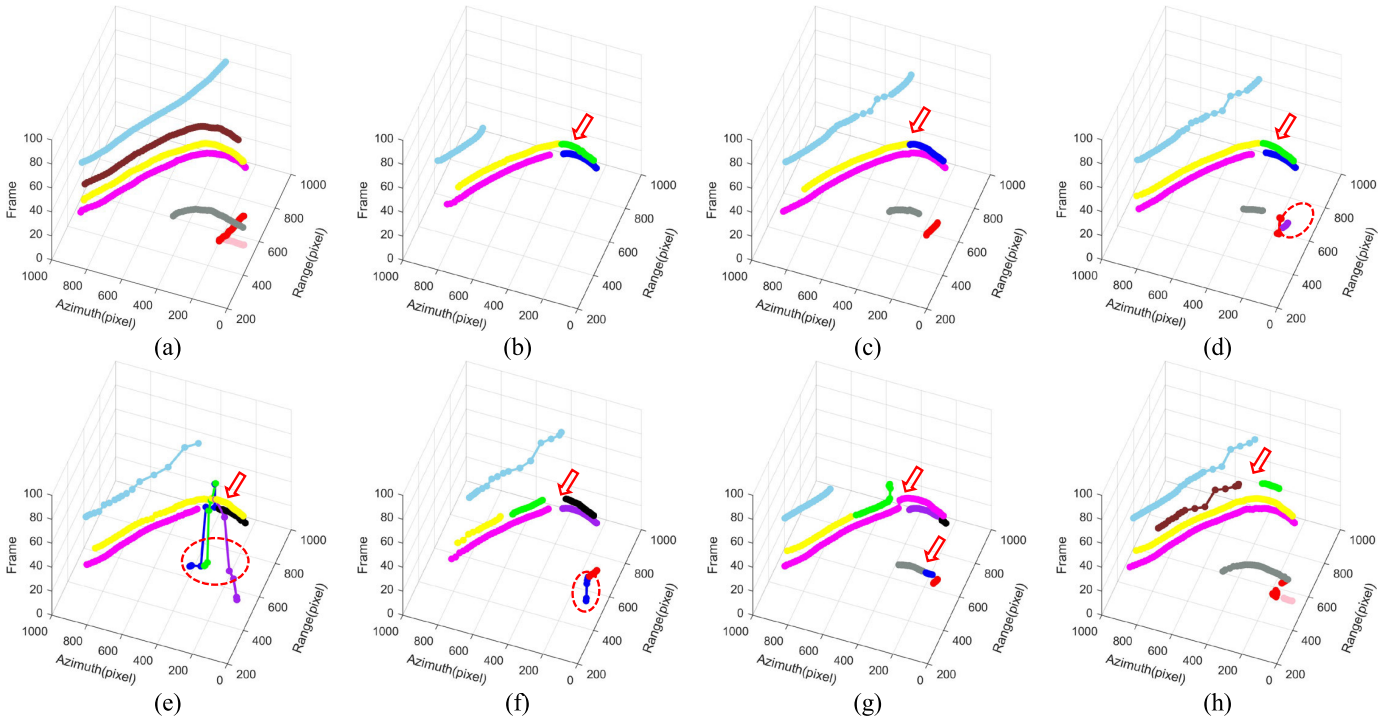



Fig. 7. Tracking trajectory with different methods in CASIC data. (a) Ground truth. (b) Bytetrack. (c) BOT-SORT. (d) OC-SORT. (e) SiamMOT. (f) FairMOT. (g) QDtrack. (h) GNN-JFL (ours). The red dashed elliptic indicates a false trajectory, and the red arrow indicates trajectory ID switching.

TABLE IV
PERFORMANCE FOR DIFFERENT PARAMETERS TO OUR GNN-JFL IN SNL DATA

| $\mu_1$ | $\mu_2$ | MOTA(%)↑ | FP↓ | FN↓ | IDSW↓ |
|---|---|---|---|---|---|
| 0.4 | 0.1 | 58.6 | 290 | 363 | 47 |
| 0.4 | 0.3 | 58.1 | 274 | 380 | 55 |
| 0.4 | 0.5 | 54.6 | 228 | 475 | 66 |
| 0.5 | 0.1 | **60.7** | 237 | 385 | 43 |
| 0.5 | 0.3 | 59.9 | 216 | 409 | 54 |
| 0.5 | 0.5 | 56.8 | 151 | 512 | 69 |
| 0.6 | 0.1 | 58.5 | 190 | 454 | 36 |
| 0.6 | 0.3 | 57.7 | 164 | 504 | 48 |
| 0.6 | 0.5 | 53.6 | 121 | 608 | 57 |

TABLE V
PERFORMANCE FOR GRADUAL ADDITION OF IMPROVEMENT TO OUR GNN-JFL IN SNL DATA

| GNN(A) | GNN(M) | GAT | MOTA(%) | IDSW↓ |
|---|---|---|---|---|
| - | - | - | 41.5 | 124 |
| √ | | | 56.7 | 111 |
| √ | √ | | 60.3 | 49 |
| √ | √ | √ | **60.7** | 43 |

deterioration of extracted appearance features. Appearance-based tracking methods, such as FairMOT, SiamMOT, and QDTrack, exhibit a significant performance degradation, while the motion feature-based BOT-SORT method achieves a suboptimal performance. However, by combining motion and appearance features with graph neural associations, GNN-JFL still achieves the highest MOTA and IDF1, with the lowest missed detections, which shows its excellent migration ability and generalization.

### *E. Parameter Sensitivity Analysis*

We present a sensitivity analysis of the crucial parameters in GNN-JFL, namely, the detection threshold $\mu_1$ and tracking threshold $\mu_2$. $\mu_1$ is the confidence threshold of the detector, which determines whether the detection box is used, $\mu_2$ is the matching threshold of the Hungarian algorithm in the tracker, which determines whether the detection box is assigned an ID. Table IV shows the quantitative results of GNN-JFL with different $\mu_1$ and $\mu_2$ values. From Table IV, with the increase of $\mu_1$, many FPs can be eliminated, but it will lead to more missed detections. Also, we find that as $\mu_2$ increased, MOTA tends to decrease. This may be due to an increase in detection boxes that have not been assigned IDs. Therefore, we choose $\mu_1 = 0.5$ and $\mu_2 = 0.1$ as the final result because MOTA reaches the best.

### *F. Ablation Studies*

We tested the key components (appearance GNN, motion GNN, and GAT) to understand their impact on tracking performance. Specifically, we compare the quantitative results for the gradual addition of the three improvements to the baseline with the baseline results, as shown in Table V. The baseline of the proposed method resembles DeepSORT, which directly uses CNN to extract appearance features, uses cosine similarity to calculate the adjacency matrix, and uses the Hungarian algorithm for data association. The following will

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introduce ablation studies conducted on GNN-JFL with or without each improvement.

From Table V, we can draw conclusions as follows.

1) Tracking accuracy rises with gradual adding the three innovations. Specifically, GNN-JFL raises from the original 41.5% MOTA to 60.7% MOTA with a huge improvement of 19.2% and reduces the ID switching from 124 to 43. From the above experimental results, this proves that the proposed method is indeed very helpful in improving tracking accuracy.
2) Appearance GNN improves MOTA by 15.2% and decreases IDSW by 13. This is because the feature interaction mechanism of appearance GNN can enhance the appearance features of shadows, thereby achieving better feature matching for subsequent data association and avoiding a substantial amount of missing detections caused by association errors. In addition, it can improve the robustness of the algorithm to background occlusion problems, which is beneficial for suppressing mismatches caused by short-term occlusion. Although the use of GNN inevitably brings a certain amount of computation, we believe that it is worth considering the significant improvement in tracking accuracy.
3) Motion GNN improves MOTA by 3.6% and decreases IDSW by 62. By increasing the branches of motion GNN, the GNN-JFL's robustness to shadow appearance changes can be improved, which is beneficial for suppressing mismatches caused by short-term shadow appearance changes. The proposed method achieves appearance and motion feature learning jointly, utilizing the complementary characteristics of these two features to make the association results more accurate and greatly reduce ID switching.
4) GAT improves MOTA by 0.4% and decreases IDSW by 6. In contrast to the constraints inherent in GCN when addressing graph-related problems, wherein the capacity to adapt weightings contingent upon the attribute characteristics of neighboring nodes is absent, GAT incorporates the attention mechanism during the aggregation of features from neighboring nodes. This innovative approach empowers the network to selectively allocate attention to domain nodes exhibiting elevated affinity scores, specifically those associated with identical identifiers (IDs), thereby enhancing overall network performance. Therefore, due to the above advantages, GAT improves the tracking accuracy.

Table VI illustrates the impact of using GAT with different layers in the proposed method. Compared to CNN, which can improve the learning ability of the model by increasing the number of layers, GNN does not require many layers. There may occur oversmoothing issues if many GNN layers stacked. The results also indicate that using only three layers produces higher MOTA and less IDSW than using two or four layers.

TABLE VI
EFFECT OF NUMBER OF GAT LAYERS IN SNL DATA

| Method | MOTA(%) | IDSW↓ |
|---|---|---|
| 1-layer GAT | 56.5 | 114 |
| 2-layer GAT | 58.4 | 83 |
| 3-layer GAT | **60.7** | 43 |
| 4-layer GAT | 60.2 | 52 |

### G. Discussion

To effectively highlight the advantages of the proposed network in comparison to existing MOT methods, we conduct a detailed analysis encompassing both the structural aspects and performance metrics.

In terms of structure, the proposed network has joint feature extraction and GNN-based matching. While existing methods rely on single-feature extraction (appearance or motion) and Hungarian-based matching. In intricate tracking scenarios, the features obtained through joint feature extraction are more stable and robust than those obtained through single-feature extraction. In addition, GNN-based matching utilizes graph representation to learn the relationship between shadows and can make more accurate predictions than Hungarian-based matching.

In terms of performance, compared to the other six advanced methods, the proposed methods achieved the best tracking accuracy in both two actual Video SAR datasets. The low contrast between shadows and background in CASIC 23 data results in the deterioration of extracted appearance features. Hence, appearance-based tracking methods, such as FairMOT and SiamMOT, exhibit significant performance degradation, while GNN-JFL still achieves the highest tracking performance, which shows its excellent migration ability and generalization.

## V. CONCLUSION

In this article, we proposed a novel GNN-based method GNN-JFL to achieve data association in Video SAR multishadow tracking. Specifically, we first proposed the joint feature extraction module to extract more effective shadow feature for better data association. Then, to further improve association performance, we proposed GAT consisting of two branches, motion and appearance, to optimize the extracted features by graph relationships. Our method allows for end-to-end graph learning to guide association prediction and improve the accuracy of Video SAR shadow tracking by learning the intrinsic relationships between objects through graph learning.

In the two real Video SAR datasets (SNL and CASIC 23), experimental results demonstrated that our method achieves higher tracking accuracy against other six advanced methods and has excellent generalization ability. Among them, the MOTA of GNN-JFL in the SNL dataset is 4.2% better than the suboptimal method, and the MOTA of GNN-JFL in the CASIC23 dataset is 3.6% better than the suboptimal method.

Our method has high accuracy and excellent generalization, but there is still room for improvement in tracking efficiency. Future work will focus on improving tracking efficiency. In addition, there is still some work to be done to further improve the tracking performance in GNN-JFL. In the future,

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we will upgrade the detector part of GNN-JFL by using the multiframe relationship.

The code is available at https://github.com/wsdmw1/GNN-JFL.

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